Robot Arm: Operation and Maintenance Manual

Applies to: KUKA KR 210, ABB IRB 6700, Fanuc M-20iD. Document no. OM-ROBOT-ARM-2026.

1. Safety

Lock out and tag out all energy sources before any maintenance. Only trained personnel may work on this machine. Wear the PPE required by the plant safety rules.

2. Sensors and alarm thresholds

An alarm is raised when a value stays above the threshold for at least two consecutive samples. HIGH is a warning; HIGH_HIGH is critical.

Sensor	Location	Unit	Normal	HIGH (warning)	HIGH_HIGH (critical)
temperature	joint 2 motor	degC	approx. 50	> 72	> 85
current	joint 1 drive	A	approx. 12	> 20	> 26
vibration	base	mm/s	approx. 1.2	> 3	> 4.8

3. Alarm response

HIGH (warning): Acknowledge within 15 minutes. Review the trend of the last 2 hours and schedule an inspection within 24 hours. Follow the troubleshooting table for the sensor.

HIGH_HIGH (critical): Reduce load or stop the machine within 5 minutes and notify the shift lead. Open a corrective maintenance order. Do not restart until the root cause is found.

COMM_FAULT: Sensor readings are invalid (quality 'bad'). Check the sensor cable, connector and I/O module. If the fault lasts longer than 4 hours, take manual readings every hour.

4. Troubleshooting

4.1 High temperature (joint 2 motor, degC)

Probable causes:

- Duty cycle too high for the programmed motion
- Payload above the rated value or wrong payload data
- Joint brake not fully releasing

Corrective actions:

1. Check the payload data in the controller matches the real tool and part.
2. Reduce the speed override to 70% and add dwell time in the program.
3. Test the joint 2 brake release; replace the brake (SP-ROB-BRK-J2) if it drags.

4.2 High current (joint 1 drive, A)

Probable causes:

- Collision or contact with fixtures
- Gearbox wear on joint 1
- Damaged motor power cable

Corrective actions:

1. Check the collision log in the controller.
2. Check joint 1 gearbox backlash: max 0.1 deg. Change gearbox oil every 20,000 hours.
3. Inspect the dress pack and motor cables for wear.

4.3 High vibration (base, mm/s)

Probable causes:

- Loose base anchor bolts
- Gearbox backlash
- Resonance from high acceleration settings

Corrective actions:

1. Re-torque base anchor bolts to 300 Nm.
2. Measure backlash on all axes (see current troubleshooting).
3. Reduce acceleration in the motion profile by 20%.

5. Maintenance schedule

Interval	Task
Every 30 days	Check dress pack and cable wear
Every 6 months	Check base anchor bolt torque (300 Nm)
Every 12 months	Replace controller backup battery
Every 20,000 h	Change gearbox oil on all axes